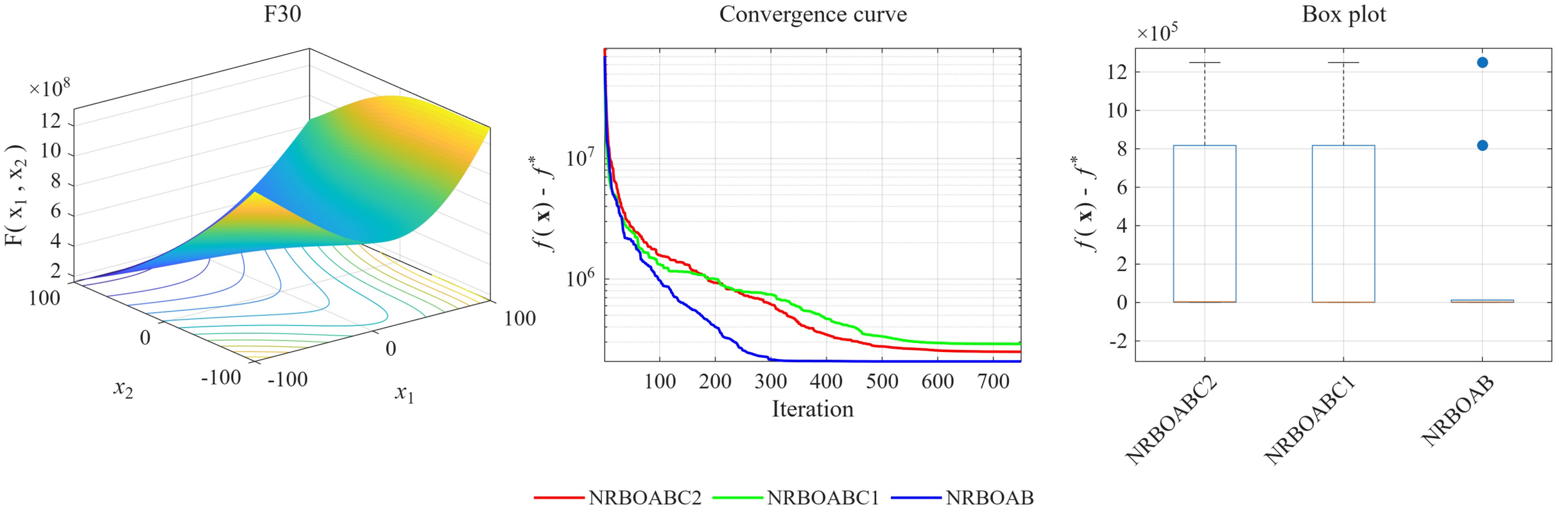

F30
×10^8
F(x1 , x2)
12
10
8
6
4
2
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^7
10^6
100
200
300
400
500
600
700
Iteration
Box plot
×10^5
f(x) - f*
12
10
8
6
4
2
0
-2
NRBOABC2
NRBOABC1
NRBOAB
NRBOABC2
NRBOABC1
NRBOAB